

College Information Assistant using Retrieval-Augmented Generation (RAG)

1.Problem Statement

Students often struggle to quickly find accurate college-related information from large documents or PDFs. Searching manually is time-consuming and inefficient. This project solves this by building an AI-powered assistant that retrieves relevant information from documents and presents it instantly.

2.Use Case Selected

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A chatbot system that helps students retrieve college-related information such as:

- Admission details
- Placement Information
- Hostel details
- Transport Information
- Library Details
- Attendance Details
- Exam Rules and Fee Structure

The system answers questions based only on uploaded college documents.

3.Knowledge base

The knowledge base consists of:

- .txt files stored in a local folder named documents
- Each file contains structured college information

The knowledge base consists of multiple text documents related to college information.

Documents used:

- Attendance Policy
 - Library Policy
 - Bus Route Information
 - Hostel Information
 - Fee Structure
- Each document contains topic-specific information used for retrieval.

4.Data Preparation

The text documents were cleaned and organized before processing. Unnecessary formatting was removed and each document was focused on a single topic.

Document titles were added to improve retrieval quality and semantic separation between topics.

Data Preparation Steps

1. Collected college-related text files
2. Stored them in a folder named documents
3. Loaded files using TextLoader from LangChain
4. Combined all documents into a single list
5. Converted raw text into structured document objects

5.Chunking Strategy

The documents were divided into smaller chunks using RecursiveCharacterTextSplitter.

Chunk Size: 300

Chunk Overlap: 50

Overlapping chunks were used in this. A chunk overlap of 50 characters was applied to preserve continuity between adjacent chunks and improve retrieval quality.

Chunking improves retrieval efficiency by allowing the retriever to search smaller and more focused sections of the documents. Chunk overlap helps preserve context between adjacent chunks. Eg: Students must maintain a minimum of 75% attendance to appear for university examinations. Students below 65% attendance are not eligible for semester exams.

6.Embedding Model used

The project uses the HuggingFace embedding model:

sentence-transformers/all-MiniLM-L6-v2

This embedding model converts text into vector representations that support semantic similarity search.

7.Vector Store Used

ChromaDB was used as the vector database for storing embeddings.

ChromaDB provides efficient semantic search and easy integration with LangChain.

8.Retrieval Process

When the user enters a question, the query is converted into embeddings. The retriever compares the query embeddings with stored document embeddings and retrieves the most relevant chunks. The retrieved context is displayed as the output.

The retriever was configured to retrieve 1 most relevant chunk for each user query. This reduced repeated or unrelated outputs and improved retrieval clarity. The user question is converted into embeddings and compared with the document embeddings stored in the vector database. The most similar chunk is retrieved. The retrieved chunk is given to the GenAI model as context so that the model can generate an answer based on the retrieved information.

9.Tools and Technologies Used

- Python
- LangChain
- ChromaDB
- HuggingFace Embeddings
- VS Code

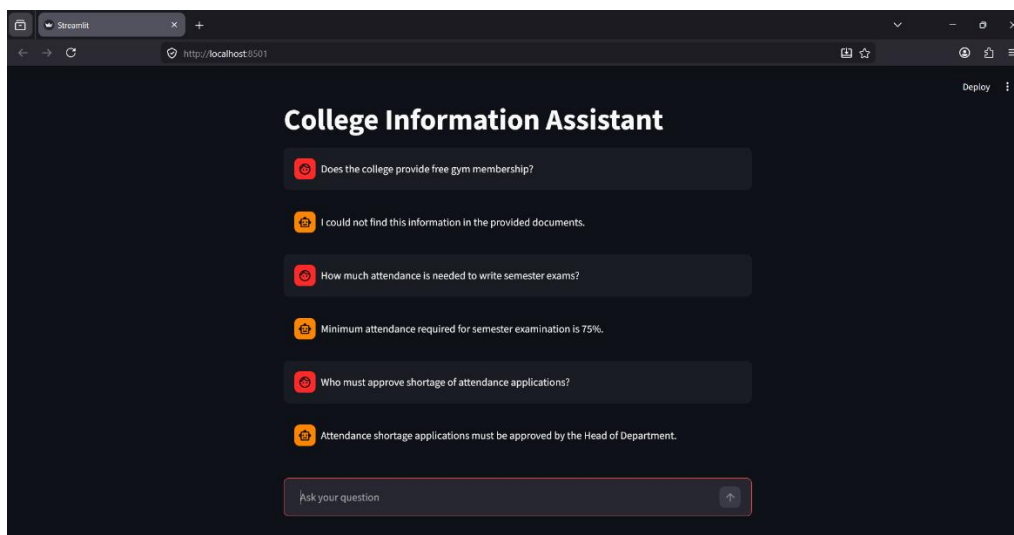
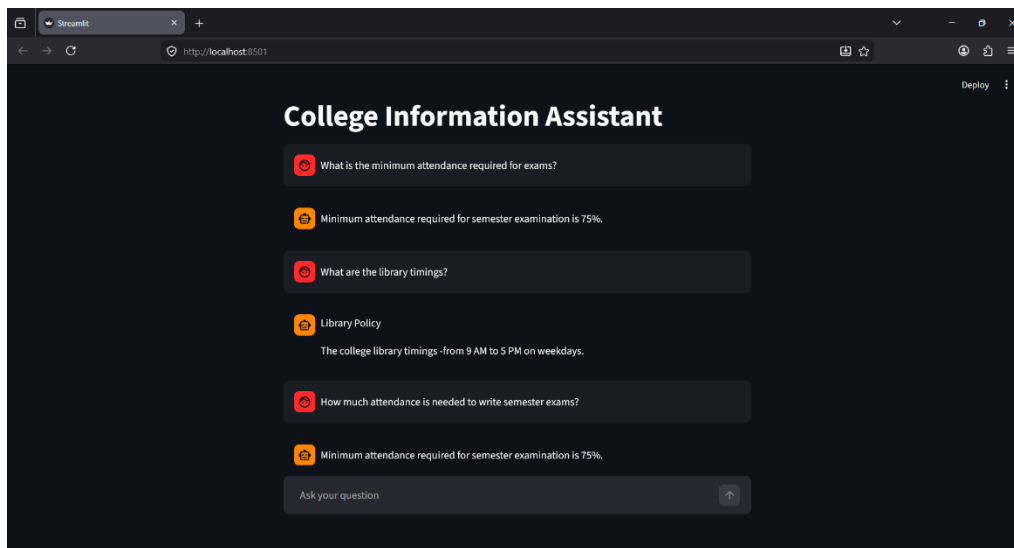
10.Prompt Used

Response logic:

- If relevant document found → return chunk content
- Else → return:

I could not find this information in the provided documents.

11.Screenshots or Sample Output



12. Test Results

Test Question	Retrieved Topic	Generated Answer	Was it Correct?
What is the minimum attendance required for exams?	Attendance Policy	Minimum attendance required is 75% for exams.	Yes
What are the library timings?	Library Policy	The library is open from 8 AM to 6 PM.	Yes
How much attendance is needed to appear for exams?	Attendance Policy	Minimum attendance required is 75% for exams.	Yes
Does the college provide free gym membership?	Not Found / Irrelevant	I could not find this information in the provided documents.	Yes
Who approves attendance shortage applications?	Attendance Policy	Attendance shortage applications must be approved by the Head of Department.	Yes

13. Challenges Faced

Initially, unrelated chunks were retrieved because multiple documents contained semantically similar terms. This issue was improved by adding clear topic titles and organizing documents more effectively.

14. Possible Improvements

- Integrate LLM (OpenAI / LLaMA) for better answers
- Improve semantic filtering using reranking models
- Add PDF support instead of only .txt
- Improve UI with chat memory summarization

15.Conclusion

This project successfully demonstrates the working of a Retrieval-Augmented Generation system using a custom knowledge base. The application performs document retrieval using embeddings and vector search, providing relevant information based on user queries.

Reflection Questions

1. Why is RAG useful for GenAI applications?

RAG is useful because it gives more accurate and context-based answers. Instead of relying only on what the model already knows, it first fetches relevant information from external documents and then generates the answer.

2. What challenges did you face while preparing the data?

One main challenge was cleaning and organizing the raw text properly. Some documents had repeated content. Preparing the data in a structured way took more time than expected.

3. How did your chunking strategy affect the quality of answers?

Chunking had a big impact on the results. If the chunks were too large, the system retrieved extra irrelevant information. If the chunks were too small, important context was missing.

4. What happened when the answer was not present in the knowledge base?

When the answer was not available in the knowledge base, the system either returned incomplete information or sometimes gave a general response based on the model's training data.

5. How can your application be improved in the future?

The application can be improved by using better embedding models for more accurate retrieval and experimenting with smarter chunking methods like semantic chunking.